

Predicting 2D Crystal Packing in Thin Films of Small Molecule Organic Materials

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The large variety of structural morphologies realized in organic semiconductors is a big challenge for the microscopic modeling of such systems.

A global computational solution is still out of reach due to prevalent molecular flexibility. However, the specific case of crystalline thin films that exhibit surface alignment of molecular π -systems for optoelectronic applications of high technological relevance, seems to be a simpler task. This study proposes an approach for the structure prediction of two-dimensional (2D) molecular layers as precursors for the three-dimensional (3D) structure of deposited crystalline thin films. Based on grid search sampling for the layer's degrees of freedom, it requires only a small number of trial structures to find complex packing motifs of layered molecular materials. It facilitates parallel screening among multiple molecular conformers, which is usually very difficult and expensive, using the latest 3D-based prediction methods. The study researches theoretically and experimentally a set of known and newly crystallized compounds of evaporable flexible molecules with interesting optoelectronic properties, predicts their packing in 2D layers, and compares them with experimentally resolved crystal structures, obtaining very good agreement in the packing of these molecules within layers. The computational costs are estimated to be several orders of magnitude lower than with 3D methods.

1. Introduction

The organic electronics market is growing rapidly due to innovative technologies including organic field-effect transistors, organic light-emitting diodes (OLEDs), and organic solar cells

(OSCs).^[1–3] Although organic electronic materials have difficulties competing with the charge-transport properties of their inorganic counterparts, they enable efficient optoelectronic devices that offer flexibility, lightweightness, electronic tunability, and sustainability and have room for further improvements.^[4,5] However, to achieve continued progress, substantial experimental efforts to study promising materials are necessary, including chemical synthesis, purification, thin film growth and characterization as well as device manufacturing for each compound of interest, which in practice is very laborious and expensive.

To support these research efforts, in silico methods for screening semiconducting candidates have gained popularity^[6,7] due to improved theoretical models and an increase in computing power. Knowing only the chemical composition allows for setting up computational schemes to predict physical properties of the molecules of interest and suggest possible high-performance materials.^[8–11] However, the key and most challenging step for organic devices

remains to be solved, namely the reliable prediction of the material's structure under real-world conditions.^[12] This is, in fact, a dual challenge because for any molecular material, it requires the prediction of the 3D molecular shape (conformation) observed in the condensed phase^[13] and the intermolecular arrangement (morphology) of the structures.^[14] Organic semiconductors show a great richness and diversity of morphologies with different levels of structural orderings of the molecules, including single crystals, thin films, interfaces, etc., which impact the performance of organic devices.^[15–20] Even for crystals, the prediction of structures is very complex in cases of high conformational flexibility, as for popular optoelectronic materials combining rigid aromatic backbones with flexible and/or bulky side groups.^[21,22] This leads to numerous energy minima on the crystal's multidimensional potential energy surface (PES), with their number growing exponentially with the system size.^[23] The task of structural prediction comes down to searching for the global minimum of the PES and other low-energy geometries likely to be found in experiments.^[24] New developments in global optimization algorithms for structure prediction at many hierarchical complexity levels have made significant progress over the past 10–15 years.^[25–28] The main workhorse for dealing with this problem in condensed phase is

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a family of crystal structure prediction (CSP) methods that enable the investigation of the most stable crystal structures.^[24] However, the progress of general CSP has slowed down for systems comprising up to 60–70 atoms and 5–6 flexible molecular dihedrals. Overcoming this limit remains challenging, even with large computing resources.^[26,29] The observation that organic semiconductors primarily display a limited array of structural motifs—most notably, herringbone and 2D brickwork patterns, with molecular layers arranged by π -stacking^[30,31] – suggests that predicting such layers could serve as a crucial intermediate step in addressing CSP challenges across a broad spectrum of organic semiconductors. This might also open the door to working with other, more exotic crystal-like morphologies and different interfaces found in organic semiconductor devices.

In this study, we present a reliable and cost-efficient workflow for predicting the thermodynamically stable arrangements of molecular layers where the structural formula of the molecules is the only necessary input information. Based on a simple grid search technique, a multi-level filtering protocol from empirical force fields to the first principles level of theory is used to identify the most stable structures from a set of trial structures. Such a reduction of the system’s dimensionality from 3D to 2D greatly simplifies the structure search, enabling studies of large sets of conformers for each molecule. Thereby this approach extends predictions of these layers to molecules with high conformational flexibility, and by minimizing the amount of computing time, the presented method establishes an efficient tool for the computational screening of promising crystalline molecular materials. When comparing the predicted 2D molecular layers with experimentally resolved 3D structures, we find for all compounds that the packing fashion of the experimental structures is among the low energy 2D polymorphs.

2. Results

2.1. Target Molecules

To establish and validate the proposed workflow, we select a set of five target molecules of different complexity for the prediction of their 2D arrangement, whose experimental crystal structures were independently determined. These molecules shown in **Table 1** include a furan-fused boron dipyrromethene (BODIPY) derivative and donor-acceptor *S,N*-heteroacenes that are of interest in organic optoelectronics because they belong to the broader class of materials that is used in active layers of bulk-heterojunction solar cells.^[32–35] Specifically, we investigate (in order of increasing molecular complexity):

- Target I – (E)-2-(5'-(1,3-dithiol-2-ylidene)-5H,5'H-[2,2'-bithiophenylidene]-5-ylidene)malononitrile.^[36]
- Target II – 5,5-difluoro-2,8-di-p-tolyl-11-(trifluoromethyl)-5H-4l4,5l4-furo[2',3':4,5]pyrrolo[1,2-c]furo[2',3':4,5]pyrrolo[2,1-f][1,3,2]diazaborinin.^[37]
- Target III – 2,2'-(((4-isopropyl-4H-thieno[3,2-b]pyrrole-2,5-diyl)bis(furan-5,2-diyl))bis-(methaneylylidene))dimalononitrile.^[38,39]
- Target IV -2,2'-(((4-propyl-4H-dithieno[3,2-b:2',3'-d]pyrrole-2,6-diyl)bis-(thiophene-5,2-diyl))bis(methaneylylidene))dimalononitrile.^[38,40,41]

Table 1. Structural formulas of target molecules studied in this work. Arrows highlight the most important flexible dihedrals.

Target	Structural formula	Flexible dihedrals
(I)		no
(II)		2
(III)		5
(IV)		6
(V)		7

- Target V – 2,2'-(3,3'-diethyl-[2,2':5',2'':5'',2'':5''-quaterthiophene]-5,5'-divinyl)bis(methanelylidene)dimalononitrile.^[41–48]

The number of flexible dihedral angles increases within this set, reflecting the growing complexity. While Target I is a rather rigid molecule, the number of flexible dihedrals increases up to seven for Target V, as summarized in Table 1. It should be noted that the identification of relevant angles depends to some extent on the intuition of the researcher; however, our choice of relevant angles here follows the common practice within the CSP community.^[29,49–51]

2.2. Conformational Search

In a blind CSP challenge, the structural formula represents the sole available information about the substance, while the molecular conformation is unknown, yet very relevant for flexible molecules. Thus, CSP involves not only predicting the mutual molecular arrangement (packing) but also finding the correct molecular conformation that will correspond to the one found in the crystal. There are two primary approaches to deal with conformational flexibility when searching a target crystal structure:

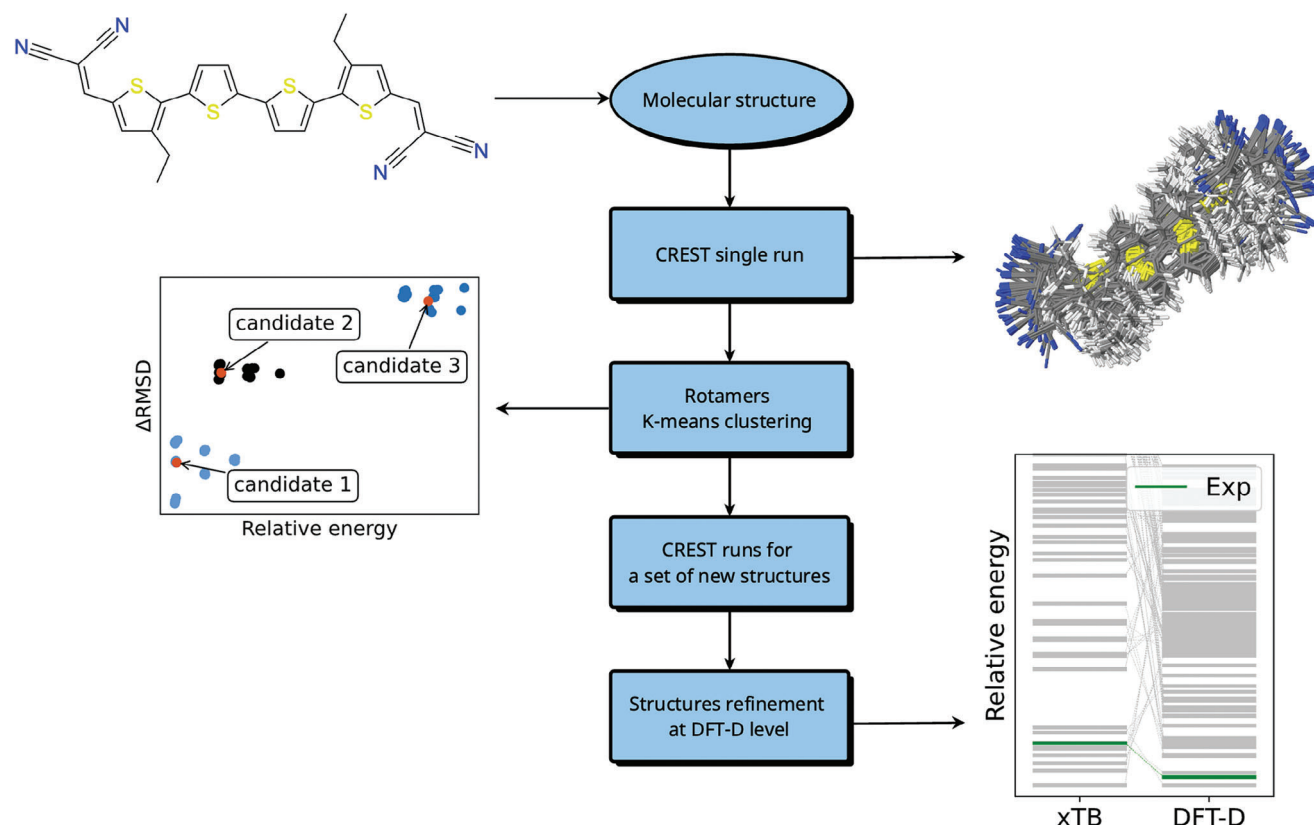


Figure 1. Schematic representation of conformational search workflow.

one is to scan the internal molecular degrees of freedom during the crystal sampling process, and another is to first explore the conformers of an individual molecule – with or without a simplified model of the crystalline environment – and then consider a subset of the most probable conformers to perform a CSP run for each of them.^[14,26] While the first approach is somewhat involved, the second one has greater popularity due to the simplicity of implementation as well as successful applications on relatively flexible molecules. Nevertheless, it also causes some difficulties, mainly with the reliability of the choice of a subset of conformers. The predicted most stable conformation of an individual molecule does not have to be observed in the crystal, where the molecule can be stabilized differently by the intermolecular environment, e.g., via crystal packing forces.^[14] Also, the elimination of human factors in conformer selection is highly desirable.^[29,52] We choose the second approach for the exploration of the conformational diversity of the target systems, due to the advantages described above, and because it avoids unnecessary search space with irrelevant structures, as encountered in the first approach.

There are many implementations of the gas-phase conformational search that can be easily adapted to fit the user's goals and objectives^[27,53] and the problem of creating the most complete and accurate energy-ranked set of conformers remains a matter of ongoing research. To date, the most successful approach is CREST,^[54,55] developed mainly by Pracht and Grimme, as it effectively studies the potential energy surface of molecules through metadynamics modeling and the use of the extended tight-binding Hamiltonian (xTB) method,^[56] which is computa-

tionally cheap and, simultaneously, much more accurate than force fields-based approaches. However, in application to two out of five target molecules of the presented paper (Target IV and Target V), single CREST runs did not find the molecular geometry of the experimental crystal structure. Based on the key idea to reduce the dependence on the initially chosen molecular geometry, we devise a slight extension of the method to generate a more complete and diverse set of rotamers and conformers. The resulting conformational search process includes several CREST runs as shown in **Figure 1**. After the first CREST run, the sampled set of conformers/rotamers is clustered by energy and geometry. Selected representative structures of these clusters serve as new starting points for subsequent CREST runs, which unveil the remaining local minima. A detailed description of each step, as well as computational details, including density functional theory (DFT) methods for the relaxation and energetic re-ranking, are provided in the Experimental Section.

The results in **Figure 2** give an overview over the set of unique conformers for Target III-V with their energies before and after DFT structure refinement and energy re-ranking. The large number of structures demonstrates the incredibly high conformational flexibility of these target molecules with many flexible dihedral angles. We also find a significant redistribution of structures for each of the target sets after refining the molecular geometry and energy with a more accurate DFT method (**Figure 2**). In particular, for the most DFT-stable conformations, we attribute this behavior to the presence of short intramolecular O or S...CCN nucleophilic-electrophilic noncovalent contacts,^[57,58] which

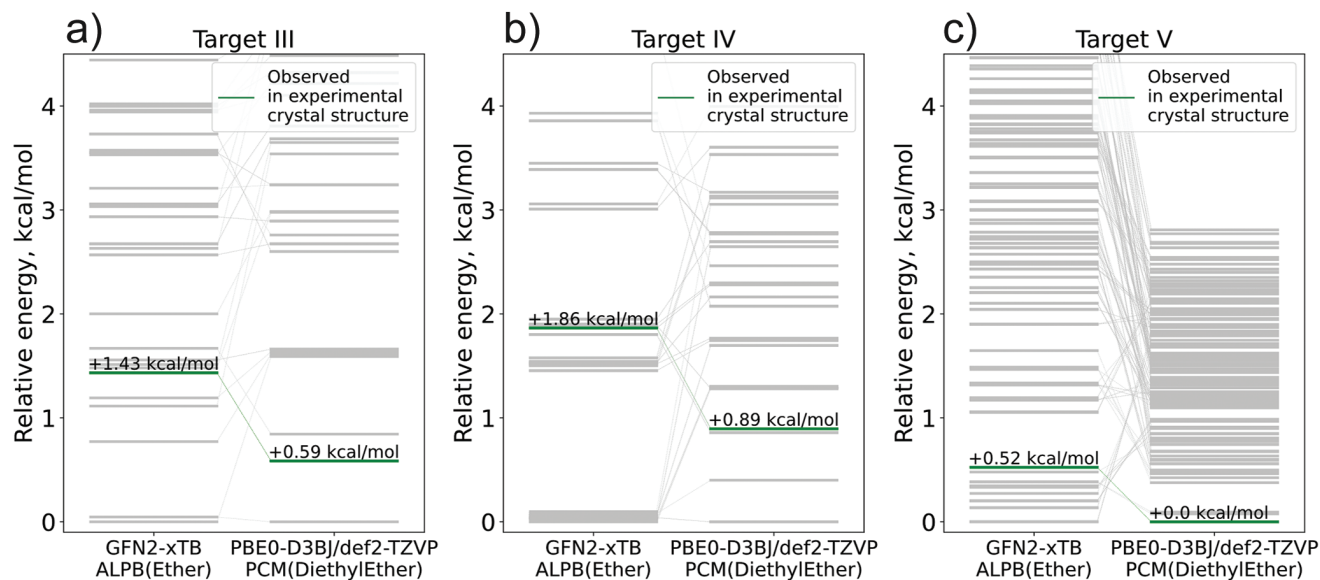


Figure 2. Resulting xTB – DFT energy ranking of final set of conformers for: a) Target III, b) Target IV, c) Target V.

further stabilize the system. At the same time, the most stable xTB-based geometries avoid that conformation, rotating the corresponding dihedral angle by 180° . The reason for this seems to be an insufficient electronic structure description of such systems by the xTB Hamiltonian, very likely leading to the failure to find the desired structures within the first CREST run. Unfortunately, there are no clear criteria for detecting the molecular geometry from an experimental crystal during conformational search, such that the choice of conformer subspace for subsequent CSP is rather determined by the amount of available computing resources. In our case, the corresponding experimental geometries were found in the top 5 conformers of the final energy ranking for Targets III–V. Hence, for demonstration purposes, this top 5 set can be identified here as a plausible subspace of conformers for which we proceed in the prediction of molecular layers. Choosing a larger number is straightforward. The complete set of conformers found for each target molecule is compiled in the Supporting Material.

2.3. Layer Structure Sampling

Formally, the molecular layer can be considered as a molecular arrangement that exhibits periodicity in two directions, where a and b are lattice constants and γ is the lattice angle. We consider only molecular geometries in which the π -systems of the molecular cores are aligned parallel to the layer, i.e., we do not consider upright standing molecules or tilted arrangements. This so-called face-down geometry yields the best performance in organic photovoltaic (OPV) devices, and we restrict our model to this orientation as there are also strategies to deliberately induce it.^[48,59–62] For concreteness, the molecular layer is defined by the centers of mass (COM) of the molecules that are equally constrained in the plane, and the initial in-plane orientation of the molecules is specified according to the principal axes of inertia of

the molecules. For a single molecule per surface unit cell, there is only one internal degree of freedom, the rotation in the plane, which is illustrated in **Figure 3a**. For each additional molecule in the 2D unit cell, we add its COM coordinates and an Euler orientation angle. The molecular layer is thus characterized by a set of degrees of freedom given as

$$\{a, b, \gamma, \varphi_1, x_2, y_2, \varphi_2, \dots, x_n, y_n, \phi_n\} \quad (1)$$

where x_i, y_i , and ϕ_i are the COM coordinates and the Euler angle, respectively, of the i^{th} -molecule in the 2D asymmetric unit. n is the total number of molecules.

In this work, we adapt the grid search method previously developed for 3D crystals,^[63,64] in which for each degree of freedom a range of possible values between $D_{n,\min}$ and $D_{n,\max}$ is determined. This interval is divided into $|D_{n,\max} - D_{n,\min}|/\delta D_n$ parts determined by the specific discretization parameter δD_n (i.e., step size), forming a set of $|D_{n,\max} - D_{n,\min}|/\delta D_n + 1$ grid points. The Cartesian product of these is called the search grid

$$G = D_a \otimes D_b \otimes D_\gamma \otimes D_{\varphi_1} \otimes D_{x_2} \otimes D_{y_2} \otimes D_{\varphi_2} \cdots \otimes D_{x_n} \otimes D_{y_n} \otimes D_{\phi_n} \quad (2)$$

which can be understood as a tensor of rank $1 + 3n$. Thus, to find a structural motif that is possibly observed in experiments, one needs to explore the low-energy minima in this high-dimensional configuration space, whose complexity increases strongly with number of molecules n in the 2D asymmetric unit and with decreasing step sizes δD_n . While each point in this grid corresponds to a certain structure of the layer, only a sufficiently dense grid with small δD_n allows sampling all degrees of freedom and covering all energetic minima.

Another important feature of the low symmetric molecules considered here is the possibility of 2D chirality for their packing in 2D layers.^[65,66] As illustrated in **Figure 3b**, in addition to the above tensorial representation, we consider that the two faces of any molecule can point either up or down. The chiral packing

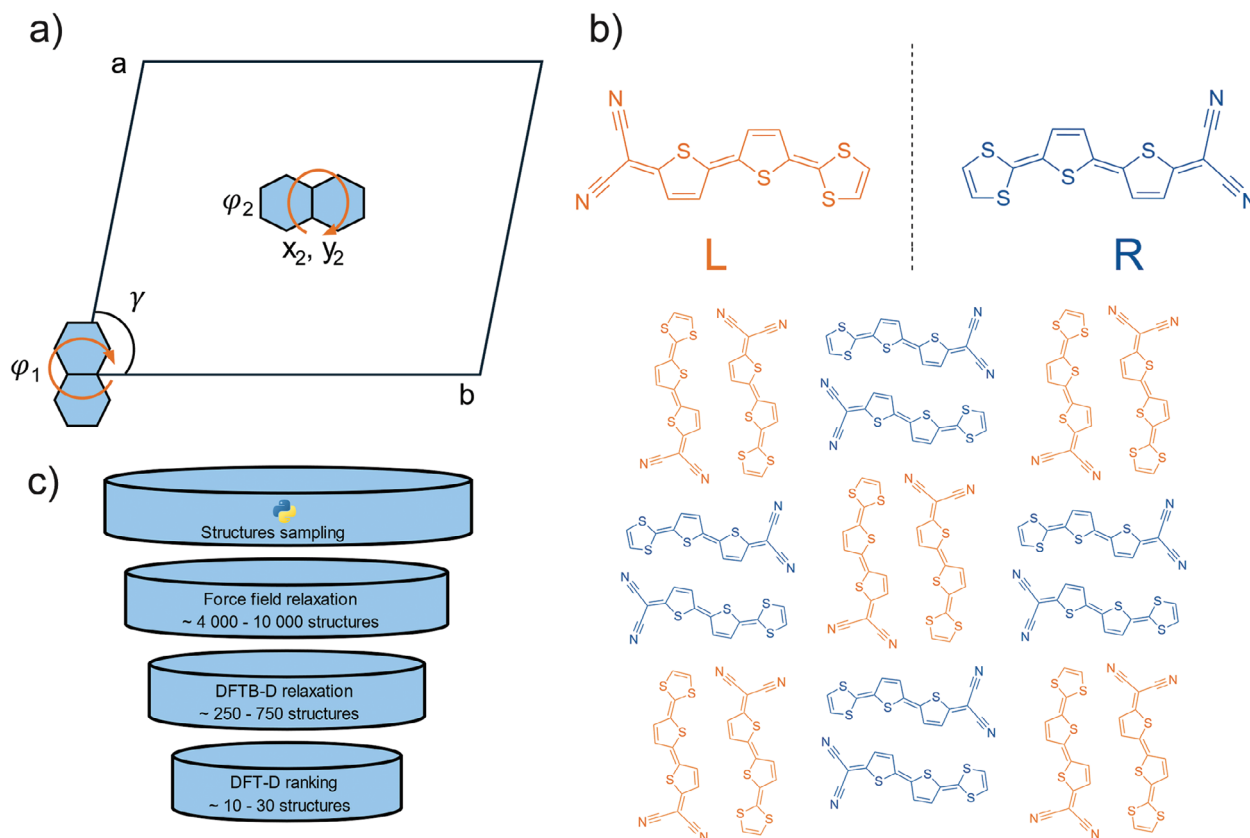


Figure 3. a) Schematic description of layer degrees of freedom with two independent molecules in a unit cell. b) Schematic representation of chiral packing in 2D for Target I. c) Layout of the multi-level filtering scheme for selecting the most stable layer structures with average number of structures at each step.

is described by the second Euler angle of the molecules in the layer, which takes only two values: 0° and 180° (up and down) and is additionally included in the general framework. Therefore, we combine different of such orientations: one combination for one molecule in a cell ($Z = 1$), two combinations for two ($Z = 2$), and three combinations for four molecules per unit cell ($Z = 4$) and define the grids for each one.

We note that despite an obvious disadvantage of a grid search, namely its exponential scaling with an increasing number of degrees of freedom, we find it much more efficient than other approaches in 3D as will be discussed further below. In any event, an important observation is that not all degrees of freedom discussed above are equally important, and specifically tailored sampling schemes are advantageous. For example, we find quite counterintuitively that densely sampling the values of cell parameters a , b and γ is not an efficient approach to searching for layer structures. This is because it will not create a high diversity of molecular arrangements during the relaxation of those generated structures. Instead, a more efficient approach is to reduce the step size for the Euler angle of the molecular orientation (for details, see Table S2, Supporting Information). This optimization allows us to reduce the required number of trial structures for finding an experimental structure. Overall, we find that ≈ 4000 structures for $Z = 2$ and 10 000 for $Z = 4$ searches are sufficient, which is significantly less than in the case of predicting crystals with 3D approaches.^[67]

2.4. Multi-Fidelity-Multi-Level Filtering Approach

The structures generated initially by the search grid are precursors that may still be far from true energetic minima, and it is necessary to minimize their lattice energy via geometry optimization. Ideally, ab initio/DFT methods are used for this but describing many thousands of structures is impractical at that level of theory. In this case, it is reasonable to set up and optimize a multi-level protocol to reduce the number of structures to the true minima. A multi-fidelity-multi-level protocol allows reducing the number of structures by filtering the lowest-energy structures and simultaneously increasing the accuracy of the simulation (Figure 3c). In the first step, all generated layer structures are relaxed by a generic force field in the rigid-molecule approximation. This allows obtaining dense packings of molecules and localizing them near the energy minima without loss of accuracy that may arise from a reduced accuracy in the description of the internal structure of the molecules. The unique structures within an energy window relative to the most stable ones are passed to the next level in the accuracy level hierarchy. In further steps, the remaining structures are sequentially refined via the more accurate density functional tight binding (DFTB) and the DFT method, where eventually ≈ 10 – 30 unique structures are selected as the thermodynamically most stable 2D arrangements. This refined approach ultimately reduces the total computational resources required for each target, which varies in the range of

4000–11 000 CPU hours (for details, see Table S3, Supporting Information).

2.5. Comparison with Experimental Structures

To validate the reliability of the proposed methodology, a prediction of possible layer structures for each of the target systems was carried out and compared with experimental crystal structures. Toward this end, we used the known crystal structures if available. For Targets III–V, we synthesized the molecules, grew the crystals by sublimation, and solved the structures by single-crystal X-ray diffraction. For Target III we followed the synthesis protocol in ref. [38], and for Target IV the one in ref. [68] (with the subtle variation that in the latter synthesis we used a mixture of anhydrous THF and anhydrous DMAc instead of DMF as solvent). For synthesis of Target V and details of crystallization see Experimental Section (Section S1, Supporting Information). Since the systems under study crystallize in a brickwork structural motif, 2D arrangements were extracted from the experimental structures and compared with those predicted. The overlays for all target systems are displayed in Figure 4 and include 18–20 molecules in each case. The complete set of geometries of the predicted 2D layers found for each target molecule is compiled in the Supporting Material for individual comparisons by the readers.

2.5.1. Target I

Target I crystallizes in space group P-1 with four molecules in the unit cell and two in the asymmetric unit (Figure 4a).^[36] For this reason, during the layer-structure search, the setting $Z = 4$ was imposed to find this structure. Figure 5a summarizes all identified polymorphs along with their energies and packing densities, with the latter expressed as an areal density measured in molecules per nm².

The experimental structure was reproduced as the global minimum for the predicted 2D packing of such molecules. It exhibits also the densest packing among all polymorphs. The top view in Figure 5a shows the 2D layer, in which the molecules are arranged in a rather complex 2D chiral structural motif of molecular dimers that form a herringbone fashion (cf. Figure 3b) with a relative orientation of 76° (Figure 4a).^[36]

As apparent in the top view, the predicted polymorph shows the same in-plane fashion as observed in the experiment, which is strongly stabilized by hydrogen bonds involving CN functional groups.

The experimental 3D crystal structure, however, is not a simple stack of strict 2D layers as illustrated by the side view. Inside the molecular dimers, we observe out-of-plane shifts of molecules relative to each other in the range of 0.495 Å to 1.048 Å, forming a staircase with shallow steps between the dimers, which is caused by a combination of intermolecular interactions. Interestingly, despite the presence of such out-of-plane shifts in the 3D crystal, the structural in-plane motif does not undergo noticeable rearrangements, which eventually leads to intriguing excitonic properties of this material.^[36] The energy of the most stable 2D polymorph is at least ≈ 1 kcal mol⁻¹ per molecule lower than that of all others (Figure 5a).

2.5.2. Target II

Target II also crystallizes in the space group P-1, with four molecules in the unit cell and two molecules in the asymmetric unit. However, here only two out of these four molecules belong to the same layer as shown in Figure 4b, while the structure is a brickwork packing of layers. Each molecule is packed into dimers, in which they are related to each other by an inversion center. The experimentally observed structural pattern in 2D layers was found at the top 4 ranked polymorph in the $Z = 2$ search, which is 0.51 kcal/mol above the top structure (Figure 5b), which is within the accuracy of DFT-D energy calculations.^[69,70] The deviations from the experimental structure might be also influenced by the presence of stacking in the 3D structure, as well as by the other surrounding bulk CF₃ groups (Figure 4b).

2.5.3. Target III

Target III crystallizes in space group P2₁/m, with two molecules in the unit cell and half a molecule in the asymmetric unit. The experimental layer was found during a search at $Z = 2$ and is slightly (≈ 0.99 kcal mol⁻¹) lower in energy than the global minimum, while at the same time also having one of the densest packings among other polymorphs (Figure 5c). The experimental structure of Target III is packed into ideal layers without out-of-plane shifts. Inside the layer, for predicted and experimental, the molecules are arranged into a simple achiral motif, in which they are related only by simple translation (Figure 4c). Despite the absence of molecular shifts in the third dimension, shifts persist within the experimental layer. The deviations of this kind are most likely due to the presence of weak intermolecular interactions (C...H, H...H contacts) and π -stacking between the layers, which shift molecular positions relative to a single layer.

2.5.4. Target IV

Target IV crystallizes in space group P2₁ with two molecules in the unit cell and one in the asymmetric unit. The experimental layer was again identified as a global minimum with high-density packing during a search at $Z = 2$ (Figure 5d). In both the experimental and predicted layers, the molecules are packed in chiral herringbone packing (Figure 4d, top view) inside the layer. Nevertheless, for the experimental layer, as for Target I, there is a tendency for both molecular shifts in the third dimension of the order of tenths of an angstrom to also affect the intralayer distances between the molecules (Figure 4d, side view).

2.5.5. Target V

Target V crystallizes in space group P-1, with one molecule in the unit cell and half a molecule in the asymmetric unit. In this case, the layer predicted in the $Z = 2$ structure search reproduces the high-density experimental layer and is 0.83 kcal/mol above the global minimum (Figure 5e). In this case, the packing of the crystal is very similar to that of the Target III crystal, with the exception that in Target V inside the layer, there are small z-direction shifts of 0.207 Å between 1D arrangements within the stacked layers (Figure 4e).

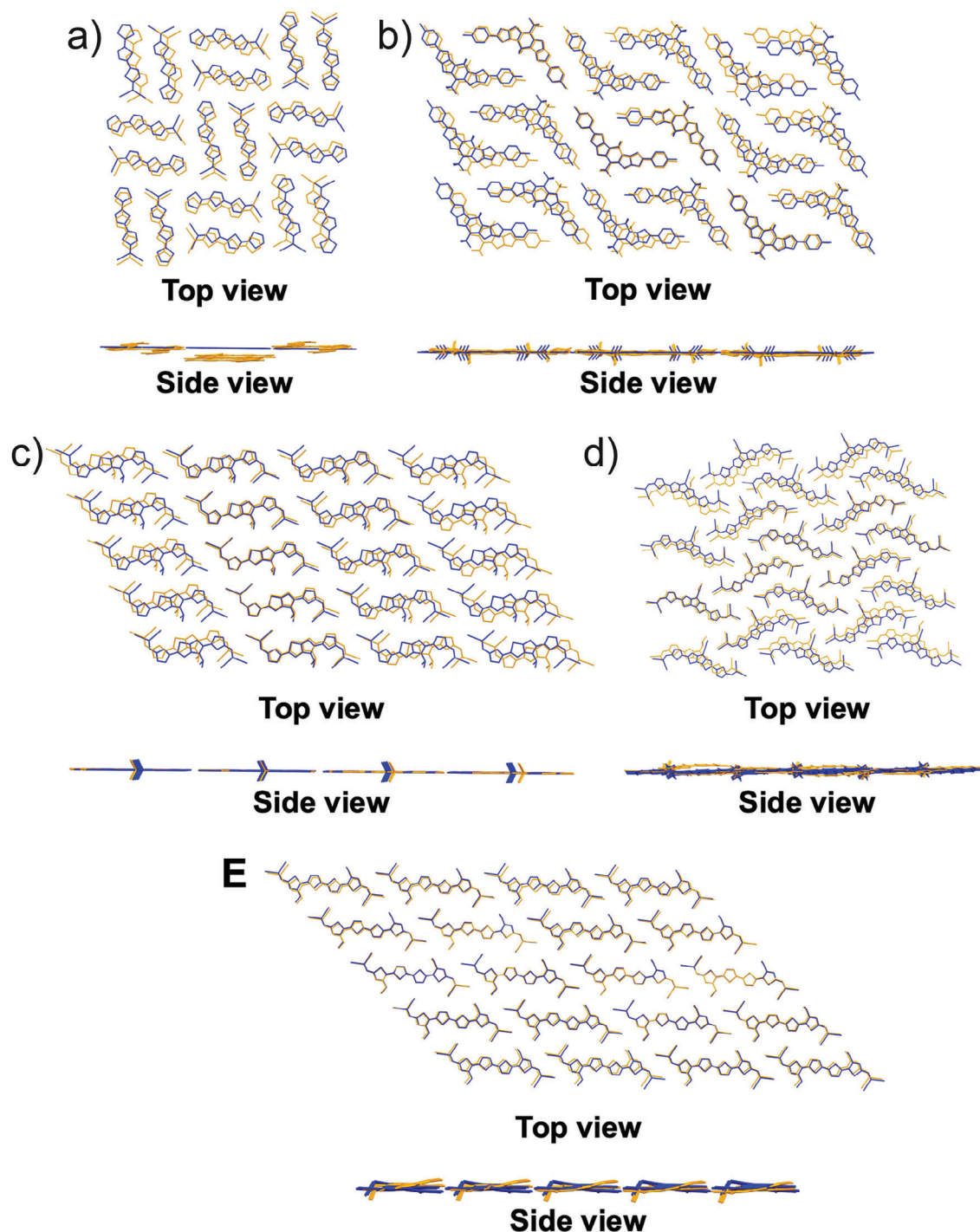


Figure 4. Structure overlay between the layers extracted from experimentally determined 3D crystal structures (orange color) and the corresponding predicted layer structures (blue color) for systems: a) Target I, b) Target II, c) Target III, d) Target IV, e) Target V. Figures show both top view and side view.

3. Discussion

Here we have investigated the relationship between 2D packing and layered 3D packing of organic photovoltaic materials, which has rarely been explored because of the flexibility of these molecules. Molecular shifts in the direction perpendicular to the

2D layers, driven by intermolecular interactions in the crystalline films, are relevant in this context and occur in a few of our target materials, most prominently in Target I. Surprisingly, they do not critically affect the packing in 2D, which may explain the success of the present approach in which such vertical shifts are not explicitly accounted for.

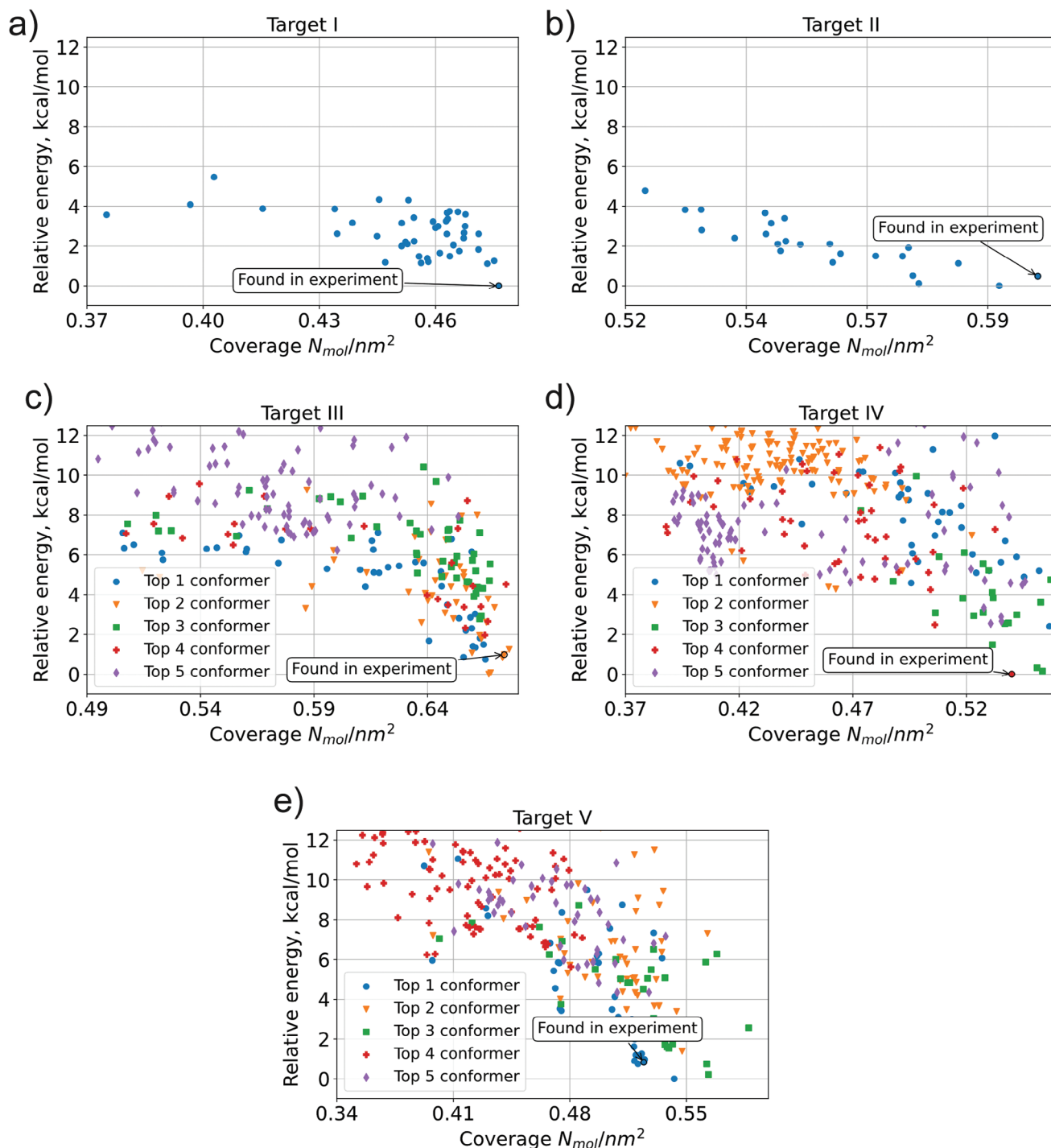


Figure 5. Energy-density scatter plot (polymorphic landscape) of predicted 2D layer polymorphs for systems: a) Target I, b) Target II, c) Target III, d) Target IV, e) Target V.

In general, the predicted polymorphic landscapes of molecular layers in Figure 5 indicate also a great richness in packing and diversity for each target molecule (see Supporting Material for all structures). Only small differences in lattice energies are observed between the polymorphs, despite large differences in molecular (conformational) structure and in structural pack-

ing motifs. This emphasizes the importance of striving for the most complete search for all possible low-energy structures, since these are the most likely ones to be observed in experiments. On the other hand, experimental structures are not guaranteed to be the thermodynamically most stable ones, but only to occur in a reasonable range of low-energy polymorphs, leading to the

postulate of kinetic control of crystallization.^[26,69,71–74] Thus, screening of promising candidates based on our method includes on purpose a certain number of possible low-energy structures (in a window of the order of 1 kcal/mol). In the present study, for two out of five cases, the experimental structures are found to be the global minimum in 2D layers, while for the other targets they are very close to them, with energy differences comparable to the DFT accuracy range.

As another result of our study, we find that an important role in the prediction is played here by the conformational polymorphism of the molecules in the layers. The predicted structures confirm that the most stable conformation in the individual-molecule calculation does not necessarily form the most stable crystal packings.^[13,14] Indeed, most of them are reshuffled in the final landscape of polymorphs. This emphasizes the importance of both an accurate conformational search and a sufficiently large set of conformers that is considered for the prediction of crystal structures.

For all five studied materials, the experimental 2D layer arrangement exhibits a high 2D packing density when compared to the entirety of all predicted 2D polymorphs. This renders these materials interesting for application in OPV devices because dense packing is indicative of strong intermolecular interactions and correlated to a flat orientation of the molecules within layers, and consequently leads to good packing in 3D. Moreover, it has been shown that the photo-oxidative stability of OSC materials tends to increase with thin film density.^[15] A higher molecular packing density can also increase the dielectric constant ϵ_r and thus the fill factor and the efficiency of an OSC.^[75]

In conclusion, starting solely from the molecular structure, the developed workflow is capable of predicting the experimental 2D arrangement of flexible OPV molecules among the lowest energy polymorphs. This renders the approach an efficient tool to simulate structural patterns of OPV molecules that tend to arrange face-down in a layered fashion in thin films.

The proposed approach not only predicts the periodic ordering of such systems with good accuracy, but also does it very economically in terms of computational resources. Each target requires ≈ 4000 – $11\,000$ CPU hours (cf. Supporting Information), which is one to two orders of magnitude less than what similar 3D methods require for predicting systems of comparable size and even fewer flexible dihedrals.^[26,76,77] There are possible ways of further reducing the computational costs of the approach. For example, the generation of structures can be restricted to a fixed space group, thereby further reducing the computational effort. We do not restrict our framework in this regard. As previously noted, the limited range of favorable bulk packings for OPV materials is enabling for the present approach focusing on 2D layers, which in turn allows one to incorporate a much larger degree of molecular flexibility. Given that reports on the blind prediction of systems of similar conformational flexibility are still rare, this research may be a blueprint for related approaches in lower dimensions. By predicting just one layer, it could be easier to move on to predicting 3D crystals that are complicated or impossible to predict by conventional 3D structure search methods. Also, molecular interfaces on substrates are conceivable in the future. The results presented in this work inspire confidence

that this approach could significantly enhance the search process itself and will become a powerful tool for predicting the morphology of molecular and supramolecular structures of larger complexity.

4. Experimental Section

Details on the Conformational Search Workflow: In the first step of the conformational search, the drawn molecular geometry (Figure 1. Molecular structure) was pre-optimized at the GFN2-xTB level of the theory and run through a single CREST run (Figure 1. CREST single run), also at the GFN2-xTB level. The resulting set of unique rotamers was sorted according to two parameters: the difference in energy relative to the most stable one and the root-mean-square deviation (RMSD) of atomic positions relative to the most stable one. This data set can be separated into several sets using any general-purpose method, k-means clustering in the case (Figure 1. Rotamers K-means clustering).^[78] From the resulting clusters, one, the most representative, structure (candidate) was taken, which is determined by the closest point to the cluster centroid. The number of candidates/clusters under consideration depends on the available computing resources, but the practice showed that at least three new structures are required for further search. For each new candidate, an additional CREST run was carried out (Figure 1. CREST runs for a set of new structures), so the total set of unique rotamers-conformers was established. Finally, the rotamers' set was refined by a more accurate dispersion-corrected density functional theory (DFT) method^[79–81] (PBE0-D3BJ/def2-TZVP) and ranked according to the new DFT geometries and energies (Figure 1. Structures refinement at DFT-D level). For CSP-purpose conformational search, it is highly preferable to include an implicit solvation effect, thereby roughly simulating molecular polarization in a crystalline environment. In the calculations, the analytical linearized Poisson-Boltzmann model^[82] was used during CREST search and the polarizable continuum model^[83] in DFT calculations for the diethyl ether solvent, as diethyl ether has a dielectric constant in the range of what organic semiconductors typically have.^[84,85] The search for conformational preferences of molecules was carried out using CREST ver. 2.12^[54,55] and xtb ver. 6.4.0 on default search settings.

Subsequent optimization of the geometries of the resulting rotamers was carried out at the PBE0-D3BJ/def2-TZVP level of theory using the Gaussian16^[86] software package. The hybrid functional scheme used PBE0^[87,88] with the def2-TZVP basis set,^[89] which has proven to be a very accurate way to model small organic molecules.^[90,91] The Grimme D3 dispersion correction with Becke-Johnson damping was also applied to better estimate the energy of non-bonded interactions.^[81]

Details on the Periodic Calculations: Relaxation of 2D layers was carried out using the DREIDING^[92] force field in the GULP program ver. 6.2.^[93] NBO (natural bond orbital) charges^[94] were taken from gas-phase DFT calculations of molecules and used for force field calculations. The DFTB calculations at the DFTB+D3 level of theory were performed using DFTB+ ver. 22.2^[95] with the 3ob parametrization.^[96] The convergence criteria for SCF were 10^{-6} eV, and the geometry relaxation was 10^{-3} eV Å⁻¹. The single-point DFT calculations at the PBE+D3 level of theory within the projector-augmented-wave method were carried out using VASP code ver.6.4.2.^[97–100] In the VASP simulations, the periodic-slab approach was used with replicas in z-direction separated by the large vacuum distance of 20 Å in order to minimize interlayer interaction. The calculations used a Gamma-centered k-points mesh with a number of k-points defined by the KSPACING default settings in VASP, which led in the vast majority of cases to a single k-point because of the large unit cells. A plane wave cutoff energy of 520 eV was used. To remove duplicate structures, an algorithm based on the comparison of the Niggli-reduced cells implemented in pymatgen library was used.^[101]

[CCDC 2392619, 2392620, 2392626, contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif].

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article. The code used for conformer search and layer structure predictions is available at <https://github.com/ortmann-lab/2D-molecular-packing>.

Keywords

computational screening, conformational search, crystal structure prediction, molecular electronics, optoelectronic materials, organic semiconductors

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